DATA CLEANING & PREPARATION

DecodeLabs | Data Analytics Project 1 | Batch 2026

Project Objective	Prepare a raw Excel dataset for reliable analysis by checking missing values, duplicates, and data formats.
Dataset	1200 rows × 14 columns
Tools	Python, Pandas, openpyxl, Microsoft Excel, Visual Studio Code

1. Introduction

Data cleaning is an important step before analysis because missing values, duplicate records, and inconsistent formats can affect the reliability of results. The purpose of this project was to clean and validate the supplied dataset using Python and Pandas.

2. Cleaning Process

Load dataset: Loaded the supplied Excel workbook into a Pandas DataFrame.

Check missing values: Checked null values in every column. Missing values were found in the CouponCode column.

Handle missing values: Replaced missing CouponCode entries with “No Coupon” so that the absence of a coupon is represented consistently.

Check duplicates: Checked for duplicate rows and duplicate OrderID values.

Validate dates: Checked the Date column and validated date values using date conversion and invalid-date checks.

Save cleaned data: Exported the cleaned dataset to Cleaned_Dataset.xlsx and checked the final quality results.

3. Final Data Quality Results

Quality Check	Final Result
Dataset size	1200 rows × 14 columns
Total missing values	0
Duplicate OrderIDs	0
Invalid dates	0
Missing dates	0

4. Data Types

The final dataset was checked for appropriate data types. OrderID and CustomerID were treated as text identifiers; Date was stored as a date/time type; Quantity and ItemsInCart were integers; UnitPrice and TotalPrice were numeric decimal values; and descriptive fields such as Product, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, CouponCode, and ReferralSource were text.

5. Output File

Cleaned_Dataset.xlsx — the cleaned Excel dataset saved in the project folder. Supporting screenshots of the Python code, validation output, cleaned dataset, and data types are stored in the Screenshots folder.

6. Conclusion

The dataset was cleaned and validated successfully. The final checks reported zero missing values, zero duplicate OrderIDs, zero invalid dates, and zero missing dates. The cleaned workbook is ready for the next stage of analysis, subject to any further business-specific validation required.

Prepared for DecodeLabs Data Analytics Project 1. Review the report and ensure the screenshots and cleaned workbook are included before submission.